

sfops (v1)

system overview



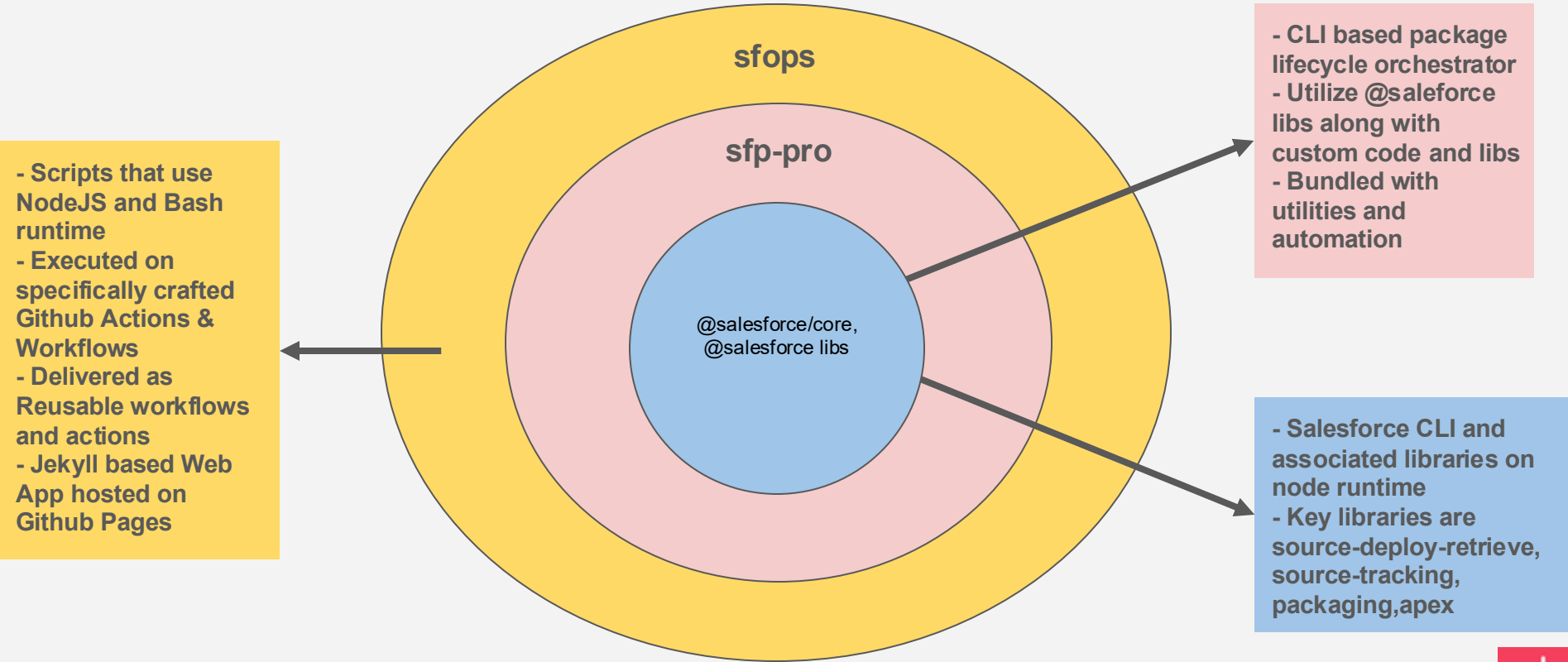
FLXBL

What is sfops?

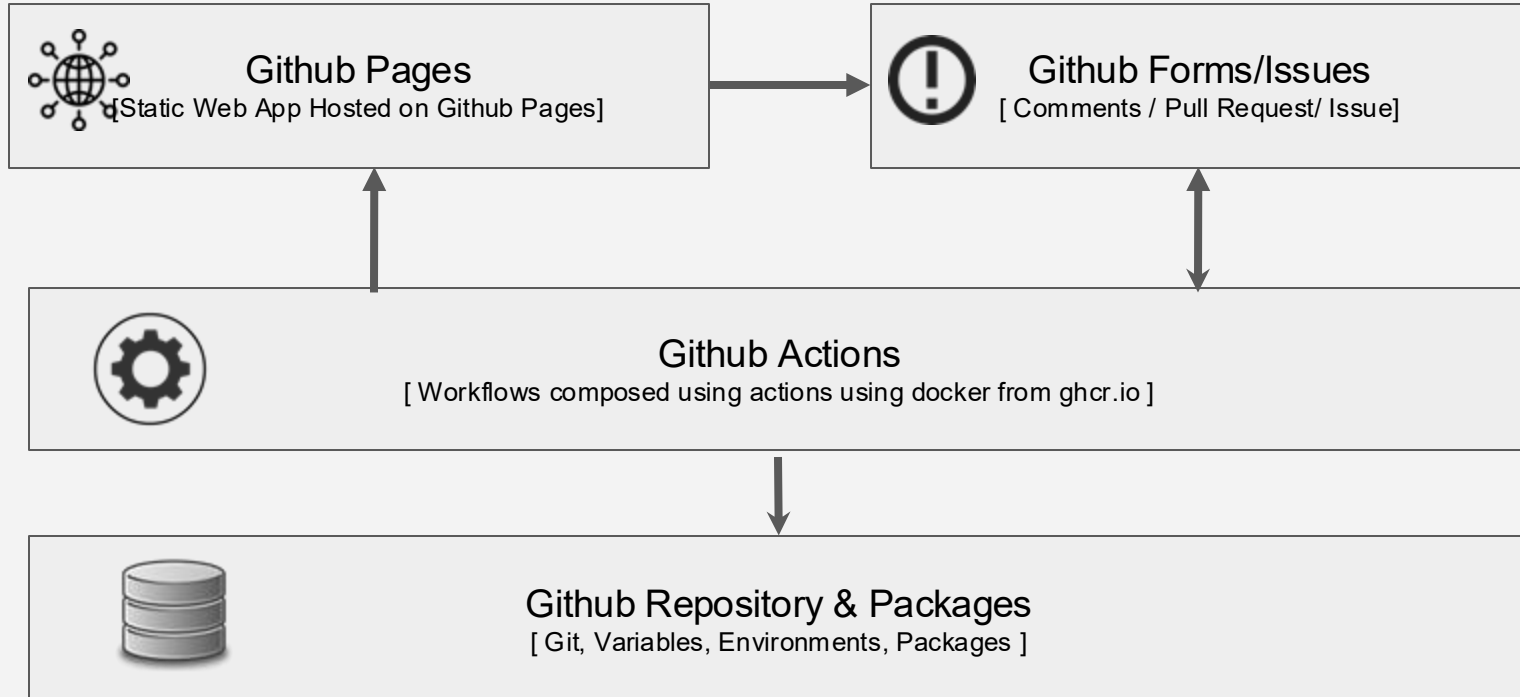
sfops is the definitive GitHub-native implementation of the Flxbl framework for Salesforce. It delivers production-ready workflows, IssueOps automation, and developer-centric tooling that is executed within your GitHub org

sfops utilize the concept of IssueOps, where the ops activities are carried out by a user by creating an issue. sfops accelerates the adoption of #flxbl model within your project and make it friendly for users who are familiar with other ClickOps solution.

The sfops stack



sfops interaction model



sfops execution layer



Github Actions

[Workflows composed using actions]

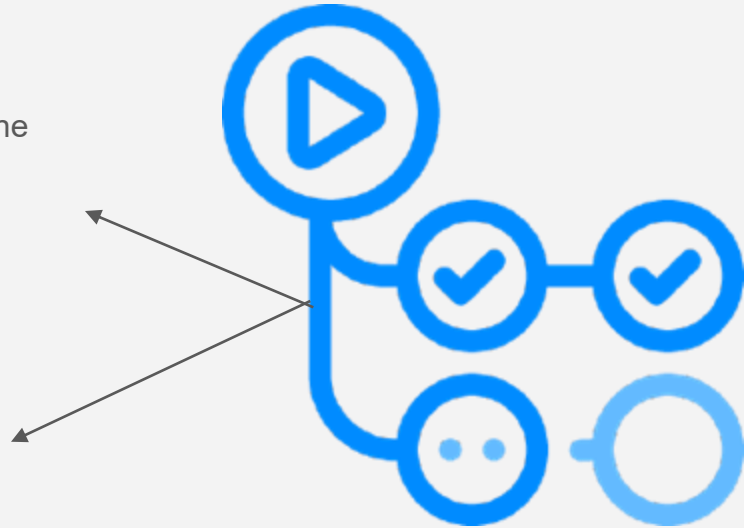


Utilizes two docker images from sfops:lite and sfops served from the org's ghcr repository



Reusable actions & workflows consumed from sfops-gh-actions repository in your Github Organisation.

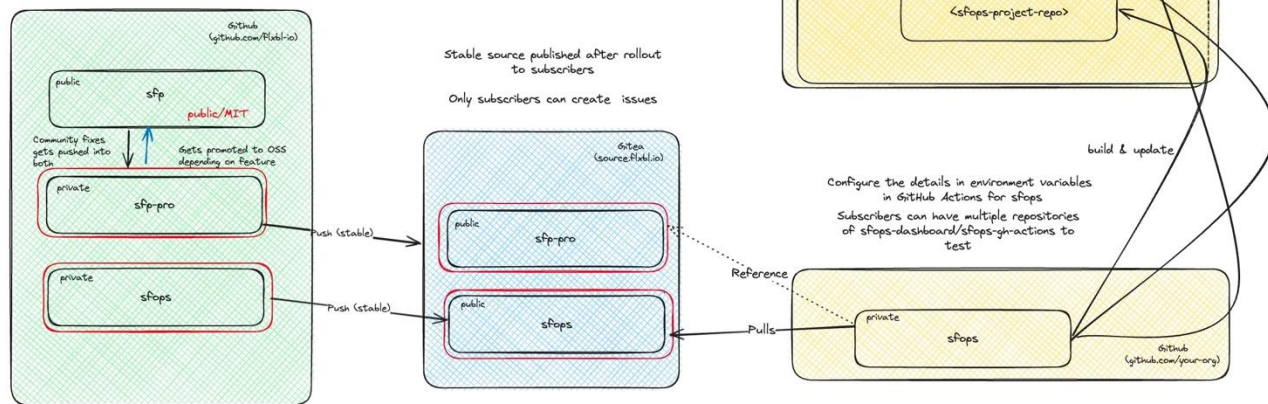
GitHub related API calls are undertaken by a custom GitHub App provisioned on the org (except for publishing packages)



Project related workflows are hosted in the repository where the code is stored

Runtime images & distribution

Self Managed sfops



sfops utilizes the `sf-pro` CLI. The `sf-pro` CLI Docker images are shipped in two variants:

sf-pro: Contains Salesforce CLI, PMD, Browserforce, Java, and `sf-pro` binaries

sf-pro-lite: Contains only `sf-pro` binaries

Docker images are published to `source.flxbl.io`, along with the source code of `sf-pro`, the images themselves, and release notes.

Self-hosted customers utilize these `sf` images as base images for generating `sfops` images (full and lite), in the forked `sfops` repository which are then utilized in GitHub Actions and Workflows.

For flxbl managed customers, `sfops` image and workflows are pushed directly by Flxbl team using a GitHub App

Decoding the flow

```
name: 'sfops - Pull Request'

on:
  pull_request:
    types: [opened, synchronize, reopened]
    branches:
      - main
      - release-*
      - release/*
  workflow_dispatch:

concurrency:
  group: ${{ github.head_ref || github.ref_name }}
  cancel-in-progress: true

jobs:
  validate-changes:
    name: 'Validate changes raised in a PR'
    uses: flxbl-io/sfops-gh-actions/.github/workflows/pr-validate.yml@main
    with:
      sfopsbot-app-id: ${{ vars.SFOPSBOT_APP_ID }}
      metrics-provider: ${{ vars.SFOPS_METRICS_PROVIDER }}
      devhub-username: ${{ vars.DEVHUB_USERNAME }}
    secrets:
      DEVHUB_SFDFX_AUTH_URL: ${{ secrets.DEVHUB_SFDFX_AUTH_URL }}
      SB_SFDFX_AUTH_URL: ${{ secrets.SB_SFDFX_AUTH_URL }}
      DATADOG_API_KEY: ${{ secrets.DATADOG_API_KEY }}
      DATADOG_HOST: ${{ secrets.DATADOG_HOST }}
      SFOPSBOT_APP_PRIVATE_KEY: ${{ secrets.SFOPSBOT_APP_PRIVATE_KEY }}
      NPM_TOKEN: ${{ secrets.GITHUB_TOKEN }}
```

1

Project workflows react to events in the project repository

2

Calls a reusable workflow or an action in the sfops-gh-actions repository

Passes variables and secrets along with context to the caller

Decoding the flow

```
name: "Validate Pull Request"
on:
  workflow_call:
    inputs:
      sfopsbot-app-id:
        description: The App id of the bot used for this workflow_call
        type: "string"
      metrics-provider:
        description: "The metrics provider to be used"
        type: "string"
      devhub-username:
        description: "The username of the devhub to be used to fetch ci sandboxes"
        type: "string"
      additional-plugins:
        description: Additional plugins used while deploying packages
        type: "string"
    secrets:
      NPM_TOKEN:
        description: "GH TOKEN to publish to NPM"
      DEVHUB_SFDX_AUTH_URL:
        description: "Devhub Auth URL"
      SB_SFDX_AUTH_URL:
        description: "Sandbox Auth URL"
      DATADOG_API_KEY:
        description: "Datadog api key to report metrics"
      DATADOG_HOST:
        description: "Datadog host to report metrics"
      SFOPSBOT_APP_PRIVATE_KEY:
        description: "Private key of the bot app used"

jobs:
  instructions:
    name: "Display instructions on every PR"
    runs-on: ubuntu-latest
    steps:
      - uses: actions/create-github-app-token@v1
        id: app-token
        with:
          app-id: ${{ inputs.sfopsbot-app-id }}
          private-key: ${{ secrets.SFOPSBOT_APP_PRIVATE_KEY }}
          owner: ${{ github.repository_owner }}

      - uses: mschick/add-pr-comment@v2
        if: ${{ vars.SFOPS_BRANCH_DEPLOY == 'true' }} && contains(github.base_ref, "release-") == false
        with:
          allow-repo-edits: false
```

3

Jobs are defined as reusable workflows get expanded during runtime and get executed

Decoding the flow

```
- name: 'Figure Impacted Release Defns'
  uses: flxbl-io/sfops-gh-actions/impactedReleaseDefnDetector@main
  with:
    pathToReleaseDefns: 'config'
    from-commit: ${ github.event.pull_request.base.sha }
    to-commit: ${ github.event.pull_request.head.sha }
```

4

Reusable actions that utilizes scripts along with wrappers around sf cli and sfp cli executes the core functionality

```
- uses: actions/create-github-app-token@v1
  id: app-token
  name: Fetch App Token
  with:
    app-id: ${ inputs.sfopsbot-app-id }
    private-key: ${ secrets.SFOPSBOT_APP_PRIVATE_KEY }
    owner: ${ github.repository_owner }
```

```
- uses: flxbl-io/sfops-gh-actions/branchDeploymentReporter@main
  name: Report Branch Status
  id: report
  with:
    status: 'validation'
    branch: ${ github.event.pull_request.head.ref }
    dashboard-repo: ${ vars.SFOPS_DASHBOARD_REPO }
    gh_token: ${ steps.app-token.outputs.token }
    issue-number: ${ github.event.pull_request.number }
```

5

Workflows create json files and submit it to sfops-dashboard-repo. A workflow on the dashboard repo then rebuilds the site which is what the user see

```
- name: Set matrix data
  id: set-matrix
  run: |
    echo "matrix=$(jq -c . < ./impacted-release-configs.json)" >> $GITHUB_OUTPUT
```

For more queries, flxbl-io.slack.com



Environments in sfops

sfops support 6 categories of environment and have associated behaviours to each of them

Development Environments

developer

- Each developer creates a unique development environment to work on work items
- A developer environment is valid only upto the max of 15 days and will automatically expire

review

- Activated when a pull request is opened against a set of defined branch
- Each Review environment is dedicated to a PR and is valid for 24 hours from when the issue is first assigned*
- A review environment can be extend for one time upto 48 hours
- A review environment is created from a snapshot environment

snapshot

- Activated on push to branch event or on pr comments
- CI / Developer Sandboxes are cloned from snapshot environments
- All snapshot environments of a branch are deployed in parallel

Environments in sfops

sfops support 6 categories of environment and have associated behaviours to each of them

Test / Release Environments

dev

- Activated on push to branch event or on pr comments
- Deploys quickbuild packages (non validated packages)
- All dev environments of a branch are deployed in parallel

test

- Activated on push to branch event or on pr comments
- Deploys build packages (validated packages)
- Can be used for jobs used daily
- All test environments of a branch are deployed in parallel

release

- Activated by specific issue template
- Prod is automatically treated as a release environment
- All release environments of a branch are deployed in parallel subject to approvals and dry run

Authenticating to Salesforce Environments

Setting up authentication to an environment is dependent on the associated approval mechanism with an environment

Required Variables	Environments created by Deployment User	Environments created elsewhere	Purpose	Where
SB_SFDX_AUTH_URL	✗	✓	Authenticate to the environment using auth URL	Environment Secret of each Github Environment
<ENVNAME>_SFDX_AUTH_URL	✗	✓	Required, if the environment is part of scheduled workflow where the job is to be executed without approvals	Repo Secret
DEVHUB_SFDX_AUTH_URL	-	-	Authenticate to DevHub and authenticate to environments created by the Deployment User	Repo Secret